

EX. 15.3

First order linear DE with variable coefficient and variable:

If an initial condition is given definitize the arbitrary constant.

$$1. \frac{dy}{dt} + 5y = 15$$

$$\frac{dy}{dt} + u(t)y = w(t)$$

$$u(t) = 5$$

$$w(t) = 15$$

Non-homogenous case.

$$y(t) = e^{-\int u(t).dt} (A + \int w e^{\int u(t).dt} dt)$$

$$= e^{-5t} (A + \int 15 e^{5t} . dt)$$

$$y(t) = e^{-5t} (A + 15 \int e^{5t} . dt)$$

$$y(t) = e^{-5t} (A + 15 \frac{e^{5t}}{5})$$

$$y(t) = e^{-5t} (A + 3e^{5t})$$

$$= Ae^{-5t} + 3e^{5t} \cdot e^{-5t}$$

$$\therefore e^{5t-5t} = 1$$

$$y(t) = Ae^{-5t} + 3(1)$$

$$e^0 = 1$$

$$2. \frac{dy}{dt} + 2ty = 0$$

$$\frac{dy}{dt} + v(t) \cdot y = w(t)$$

$$v(t) = 2t$$

$$w(t) = 0$$

Formula:

$$y(t) = Ae^{-\int v(t) \cdot dt}$$

$$= Ae^{-\int 2t \cdot dt}$$

$$= Ae^{-2 \cdot \frac{t^2}{2} \cdot dt}$$

$$= Ae^{-t^2 + C}$$

$$= Ae^{-C} \cdot e^{-t^2}$$

$$= A \cdot A' \cdot e^{-t^2}$$

$$y(t) = B \cdot e^{-t^2}$$

$$\frac{2t^2}{2}$$

$$e^{-C} = A'$$

$$A \cdot A' = B$$

$$3. \frac{dy}{dt} + 2ty = t$$

$$y(0) = \frac{3}{2}$$

$$y(t) = e^{-\int v(t) \cdot dt} (A + \int w e^{\int v(t) \cdot dt} dt)$$

$$v(t) = 2t$$

$$w(t) = t$$

$$\int v(t) \cdot dt = \int 2t \cdot dt = 2 \int t \cdot dt = 2 \cdot \frac{t^2}{2} = t^2$$

$$y(t) = e^{-t^2} (A + \int t e^{t^2} dt)$$

$$= e^{-t^2} (A + \int 2t e^{t^2} dt)$$

$$= e^{-t^2} (A + \frac{1}{2} e^{t^2})$$

$$= A e^{-t^2} + \frac{1}{2} \quad \text{G.S}$$

Condition: —

$$t = 0$$

$$y(0) = Ae^{-0^2} + 1/2$$

$$= A + 1/2$$

$$3/2 = A + 1/2$$

$$A = 3/2 - 1/2$$

$$A = 1$$

$$y(t) = 1e^{-t^2} + 1/2 \quad \checkmark \text{ Definite S.}$$

$$4) \frac{dy}{dt} + t^2 y = 5t^2 \quad y(0) = 6$$

$$v(t) = t^2$$

$$w(t) = 5t^2$$

$$\int v dt = \int t^2 dt = \frac{t^3}{3}$$

$$y(t) = e^{-\int v dt} (A + \int w e^{\int v dt} dt)$$

$$= e^{-\frac{t^3}{3}} (A + \int 5t^2 e^{\frac{t^3}{3}} dt)$$

$$= e^{-\frac{t^3}{3}} (A + 5 \int t^2 e^{\frac{t^3}{3}} dt)$$

$$= e^{-\frac{t^3}{3}} (A + 5 e^{\frac{t^3}{3}})$$

$$= A e^{-\frac{t^3}{3}} + 5 e^0 \therefore (1)$$

$$= A e^{-\frac{t^3}{3}} + 5 \quad \text{A.S}$$

$$\therefore \int e^{\frac{t^3}{3}} t^2 dt = \frac{t^3}{3} = \frac{3t^{3-1}}{3}$$

Condition:

$$= t^2$$

$$t = 0$$

$$y(0) = A e^{-\frac{(0)^3}{3}} + 5$$

$$6 = A + 5$$

$$A = 6 - 5$$

$$A = 1$$

$$y(t) = A e^{-\frac{t^3}{3}} + 5$$

$$= 1 e^{-\frac{t^3}{3}} + 5$$

$$y(t) = 1 e^{-\frac{t^3}{3}} + 5$$

D.S

$$5.2 \frac{dy}{dt} + 12y + 2e^t = 0$$

$$\frac{dy}{dt} + v(t) \cdot y = w(t)$$

$$y(0) = 6/7$$

Dividing by 2

$$\frac{dy}{dt} + 6y + e^t = 0 \quad \therefore \frac{2 \frac{dy}{dt} + 12y + 2e^t}{2}$$

$$\frac{dy}{dt} + 6y = -e^t$$

$$v(t) = 6$$

$$w(t) = -e^t$$

$$\int v dt = \int 6 dt = 6 \int dt = 6t$$

Formula:

$$\therefore \frac{1}{t} = t^0$$

$$y(t) = e^{-\int v dt} \left(A + \int w e^{\int v dt} \cdot dt \right)$$

$$= e^{-6t} \left(A + \int -e^t \cdot e^{6t} dt \right) \quad \therefore e^{t+6t}$$

$$= e^{-6t} \left(A - \int e^{7t} \cdot dt \right) \quad e^{7t}/7$$

$$= e^{-6t} \left(A - \frac{e^{7t}}{7} \right) \quad \text{P.U.A.C.P. G.I.S.}$$

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Condition:

$$t = 0$$

$$y(0) = e^{-6(0)} \left(A - \frac{e^0}{7} \right) = e^{-6t} \left(A - \frac{e^{7t}}{7} \right)$$

$$= 1 \left(A - 1/7 \right)$$

$$y(t) = e^{-6t} \left(1 - \frac{e^{7t}}{7} \right)$$

$$6/7 = A - 1/7$$

$$A = 6/7 + 1/7$$

$$= 7/7$$

$$\boxed{A = 1}$$

D.S

$$y(t) = e^{-6t} - \frac{e^t}{7}$$

$$e^{-6t+7t}$$

$$6. \frac{dy}{dt} + y = t$$

$$dy/dt + y = t$$

$$v(t) = 1$$

$$w(t) = t$$

Non-Homogenous Case

$$\int v dt = \int 1 dt = t \quad \therefore 1 = t'$$

Formula:

$$y(t) = e^{-\int v dt} \left(A + \int w e^{\int v dt} dt \right)$$

$$= e^{-t} \left(A + \int t e^t dt \right)$$

$$= e^{-t} (A + t e^t - t) \quad \therefore \int t e^t dt$$

$$= A e^{-t} + t - 1 \quad \text{Ans.} \quad = t e^t - \int (1) e^t$$

$$= t e^t - e^t$$